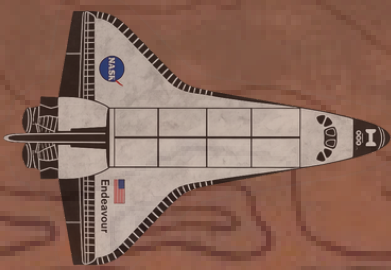


SITE ANALYSIS



Acidalia Planitia (39.8°N, 202.1°E) Location & Scale

Acidalia Planitia is a large lowland basin in Mars' northern hemisphere, near 39.8°N, 202.1°E. It is about 3,500 km wide and is one of the planet's largest topographic depressions.

STRENGTH -

- Water Ice
- Flat Terrain
- Solar Potential
- Construction Regolith

WEAKNESS-

- Extreme Cold
- Thin Atmosphere
- Dust Accumulation
- Low Gravity

OPPORTUNITIES-

- Human Settlement
- ISRU Resource Production
- Agriculture
- Scientific Research
- Mars Spaceport

THREAT-

- Dust Storms
- Radiation
- Meteoroids
- Equipment Failure
- Communication Delay



SURFACE TEMP
-63°C
(average)



PRESSURE
6.36 mbar
(0.6% of Earth)



RADIATION LEVEL
0.68 mSv/day
(average)



DUST STORMS
Global, Seasonal
(months-long)

Topography

The region lies 3,000–5,000 m below the Martian datum, with an average elevation of -3,800 m. It has a broad, gently sloping terrain covered by 2–4 km thick sediment deposits.

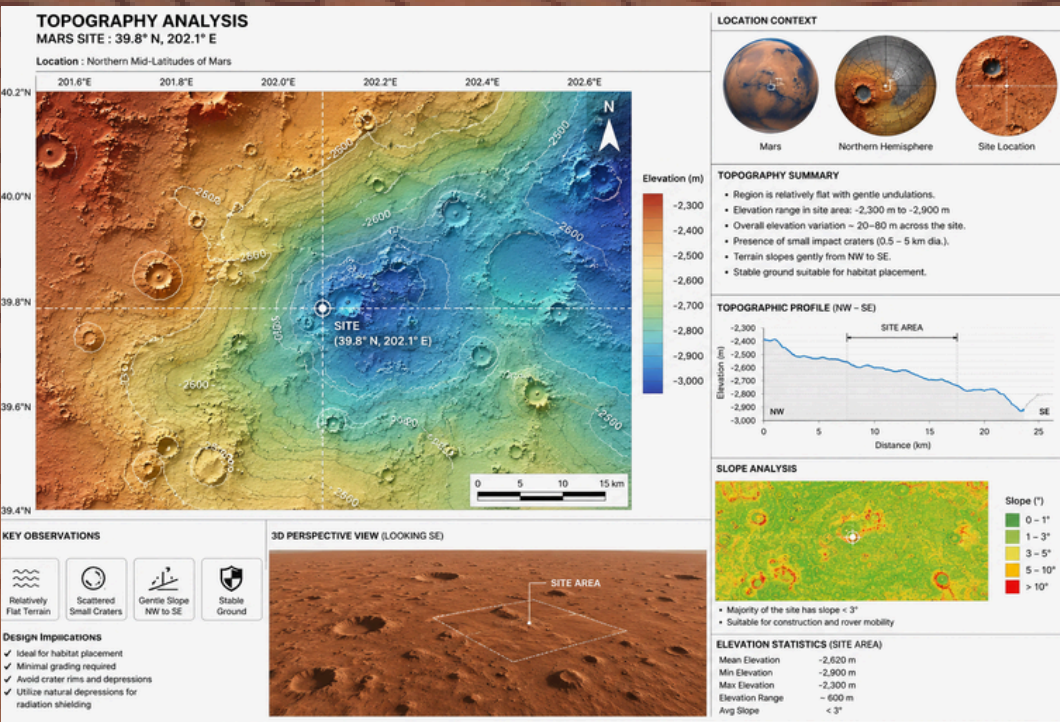


Image Source:-Gemini AI

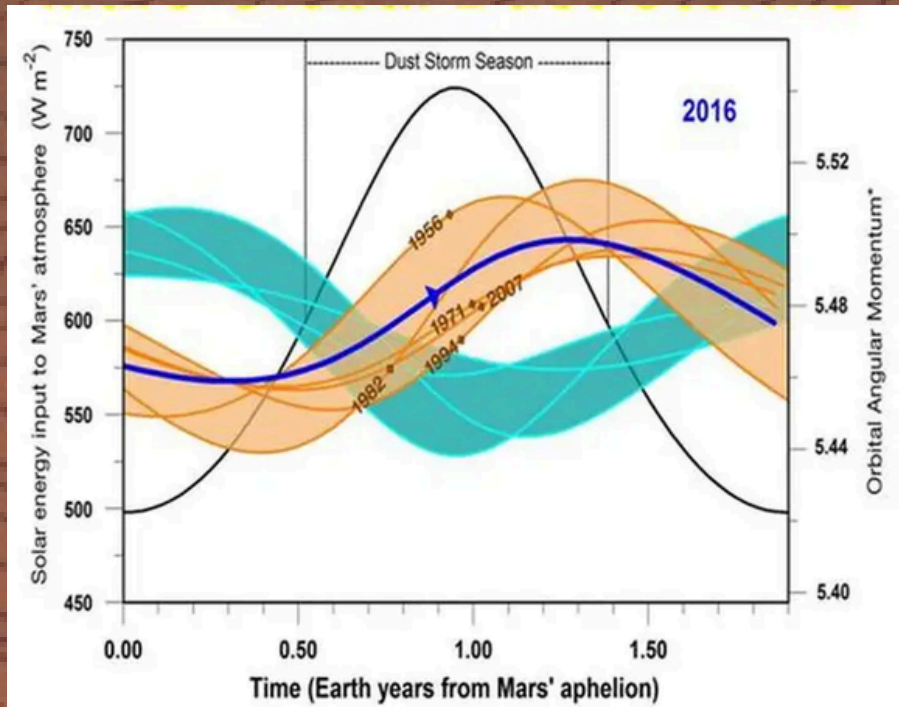
1. Earth vs. Mars: Baseline Architectural Metrics

EARTH VS MARS				
COMPARISON OF KEY ENVIRONMENTAL & PHYSICAL FACTORS				
EARTH			MARS	
9.81 m/s² (1.00 g)	GRAVITY		3.71 m/s² (0.38 g)	
1.00 AU (149.6 million km)	DISTANCE FROM SUN		1.52 AU (227.9 million km)	
365.25 days	LENGTH OF YEAR		687 Earth days	
24 hours	LENGTH OF DAY		24 h 39 m	
15 °C (Average)	AVERAGE TEMPERATURE		-60 °C (Average)	
101.3 kPa (1.00 atm)	ATMOSPHERIC PRESSURE		0.6 kPa (0.006 atm)	
78% Nitrogen 21% Oxygen 1% Other	ATMOSPHERIC COMPOSITION		95% Carbon Dioxide 2.7% Nitrogen 1.6% Argon 0.7% Other	
Abundant (liquid water)	WATER AVAILABILITY		Mostly ice (Polar caps & sub-surface ice)	
Strong (Magnetic field)	MAGNETIC FIELD		Very weak / None (No global protection)	
Low (Protected by magnetosphere)	RADIATION LEVEL		High (No global protection)	
5 - 100 km/h (Varies by location)	WIND SPEED		15 - 160 km/h (Dust storms can be higher)	
Many (Forests, plains, deserts, mountains)	LANDSCAPE		Rocky, dusty (Craters, plains, volcanoes)	
~8.1 Billion (Human population)	HUMAN PRESENCE		None (Future potential)	

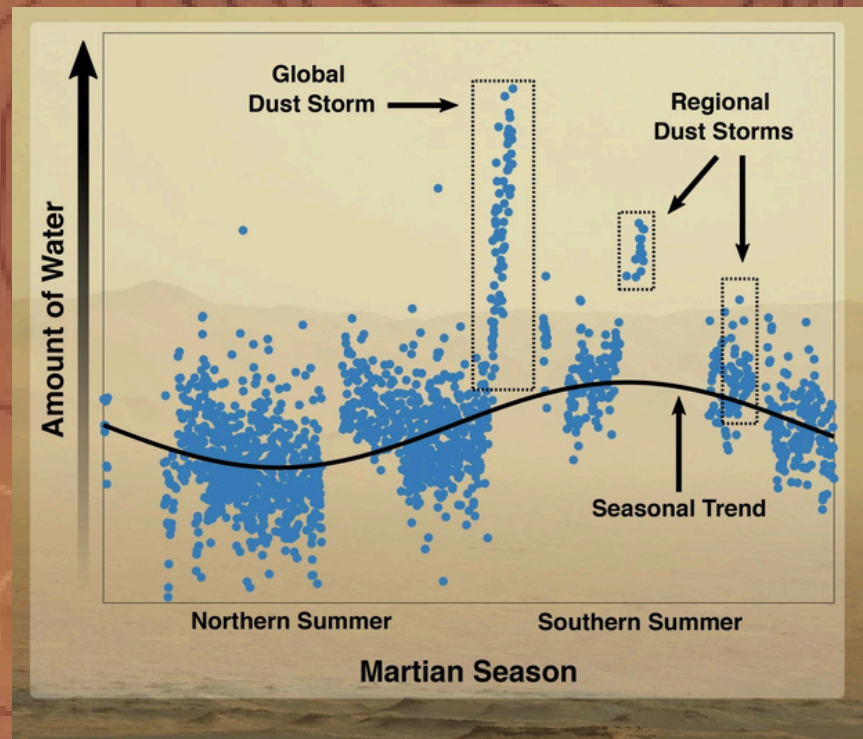
2. Weather — Seasonal Change Pattern

- **Diurnal swing:** extreme by Earth standards — surface temperatures can range roughly from near 0°C at midday in summer down to below -70°C at night, even within the same sol.
- **Atmosphere:** thin CO2-dominated (~6 mbar, <1% of Earth's sea-level pressure) — provides negligible thermal buffering, which is the main driver of the large day-night swing.
- **Precipitation:** none in the liquid sense; only frost/ice deposition possible in cold seasons.

AVERAGE MONTHLY SNOW DUST



The graph shows that global dust storms on Mars mostly occur when solar energy reaching Mars is at its peak, especially around the middle of its orbit (dust storm season). These storms become stronger as atmospheric energy and orbital conditions align, reducing visibility and significantly affecting solar power generation for habitats



Radiation Analysis

- **High radiation exposure** due to Mars' thin atmosphere and lack of a magnetic field.
- **Average radiation:** ~0.67 mSv/day.
- **Main sources:** Galactic Cosmic Rays (GCRs) and Solar Radiation.
- **Design Response:** Habitats should be buried under 2–5 m of regolith or built underground.
- **Advantage:** Presence of subsurface ice can help provide additional radiation shielding.

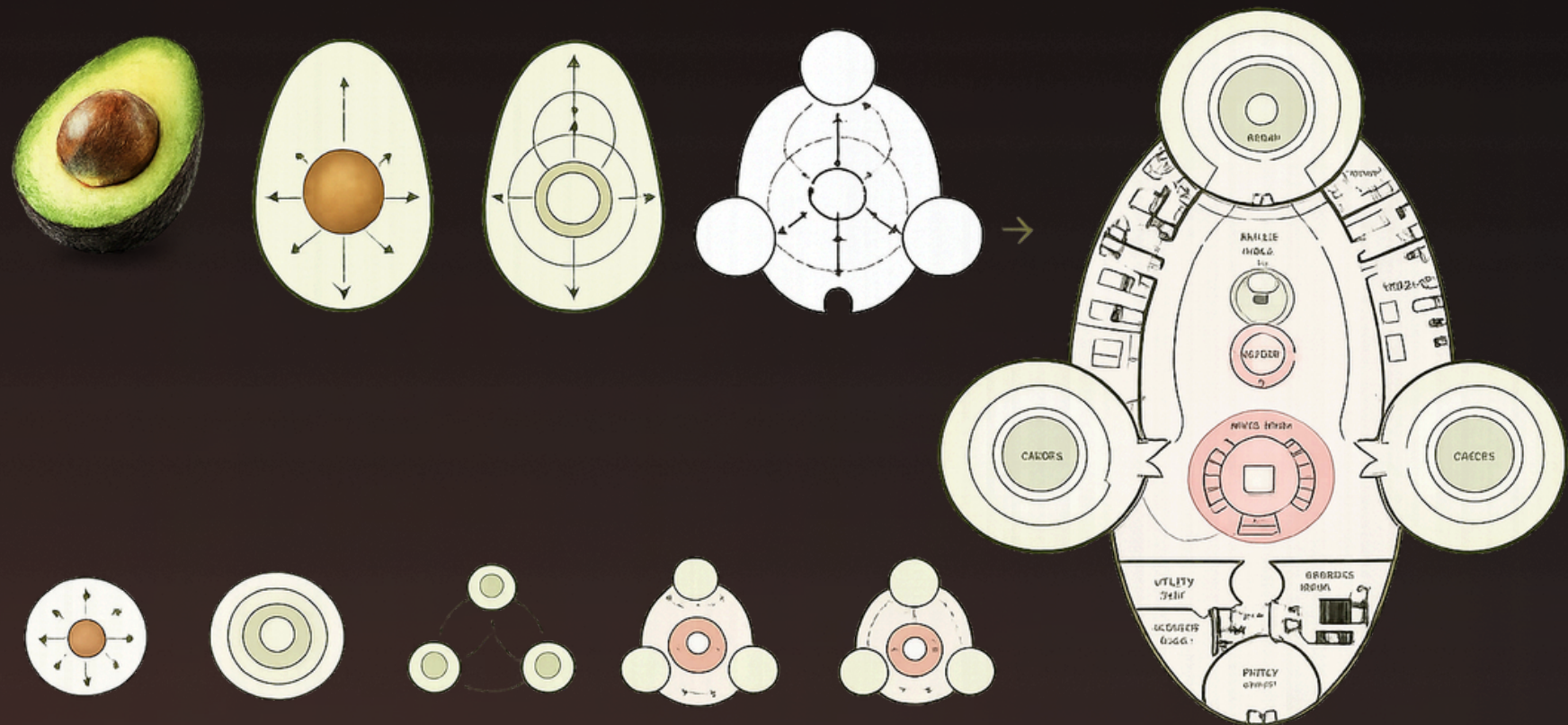


THE FIRST SEED OF LIFE

The core of this Martian habitat concept is "The Kernel," which anchors the centrally located Kitchen and Living Room. Functioning as the true heart of the home, this central bio-hub mimics the protective mechanics of a natural seed, generating vital resources like food, water filtration, and oxygen while directly supporting all surrounding residential and functional spaces.

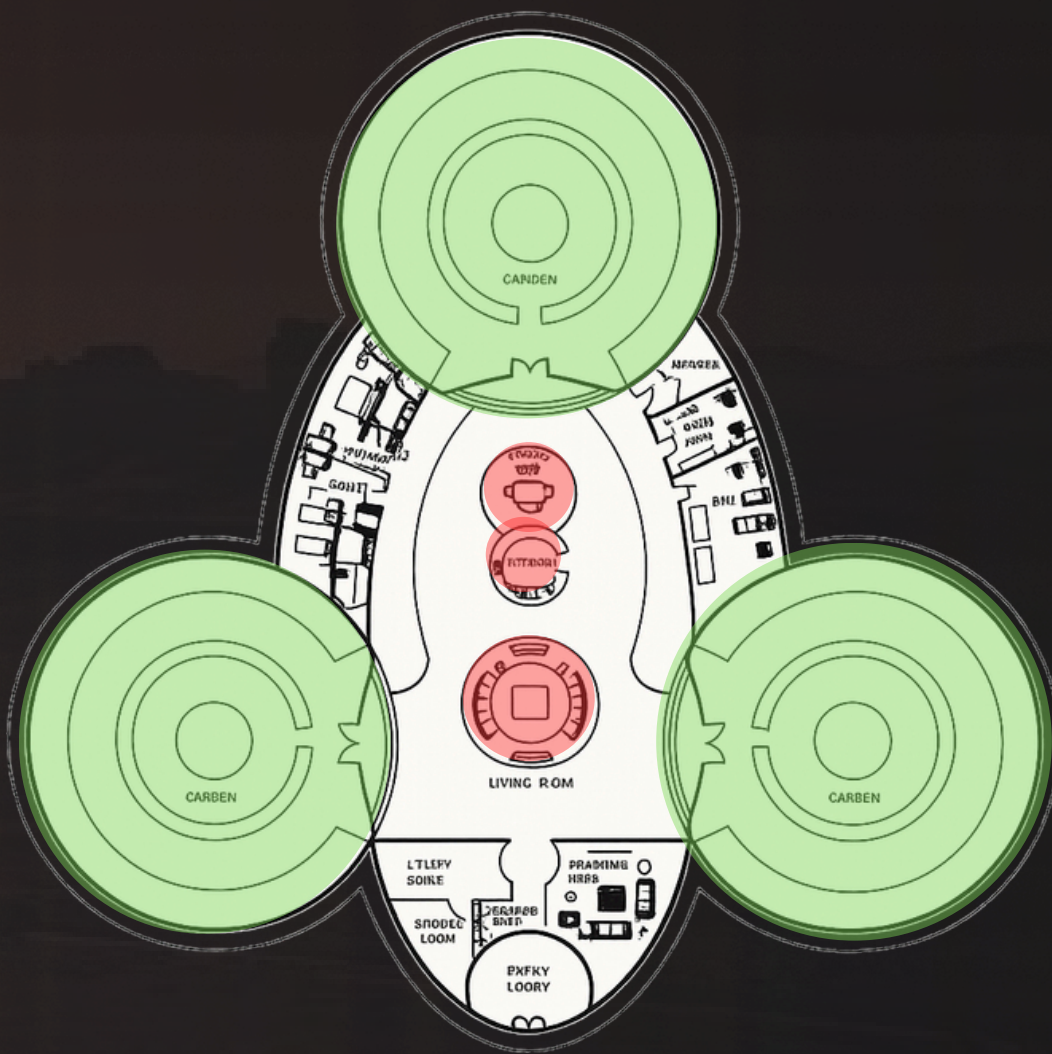
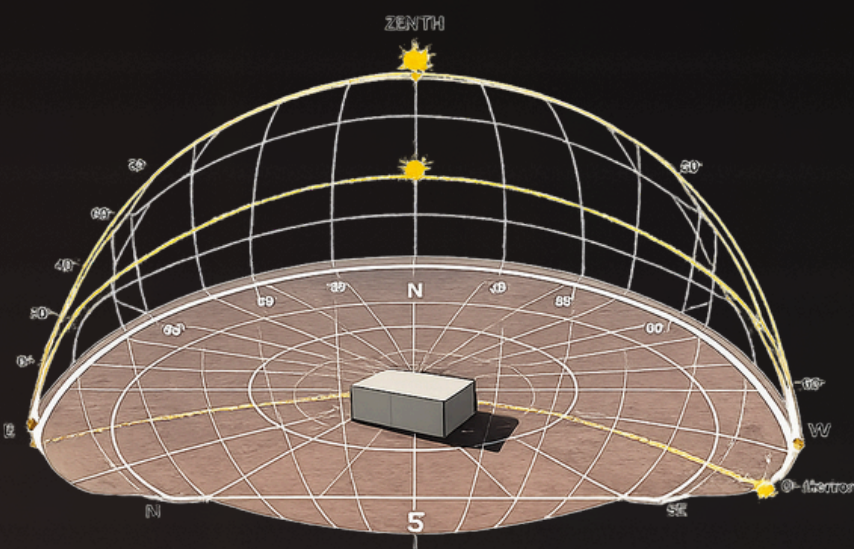
A seed is the most radical object in nature — a complete universe of potential compressed into the smallest possible form. It carries its own food, its own blueprint, its own protection. Placed in hostile ground, it does not adapt to the environment. It transforms the environment around itself. This is exactly what the first human house on Mars must do. The architecture IS the seed. The house is not built on Mars — it is planted there. And from it, life radiates outward into the red soil.

FORM DEVELOPMENT



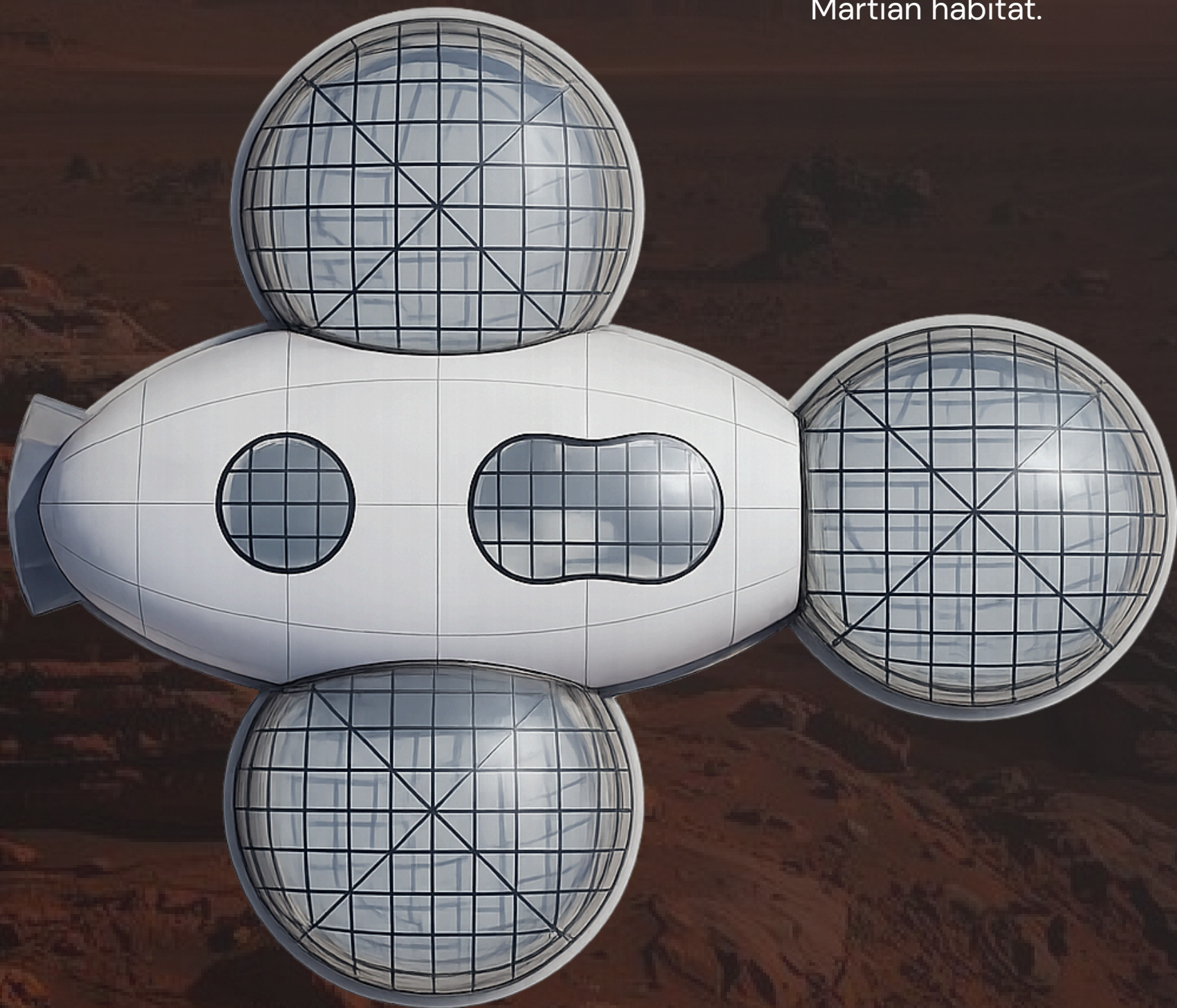
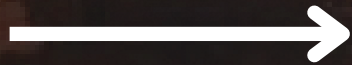
The avocado's seed inspired the central Kitchen and Living Room, the heart of the home that supports all spaces. The organic form and green domes provide protection, food, and oxygen, creating a sustainable Martian habitat.

SUN PATH

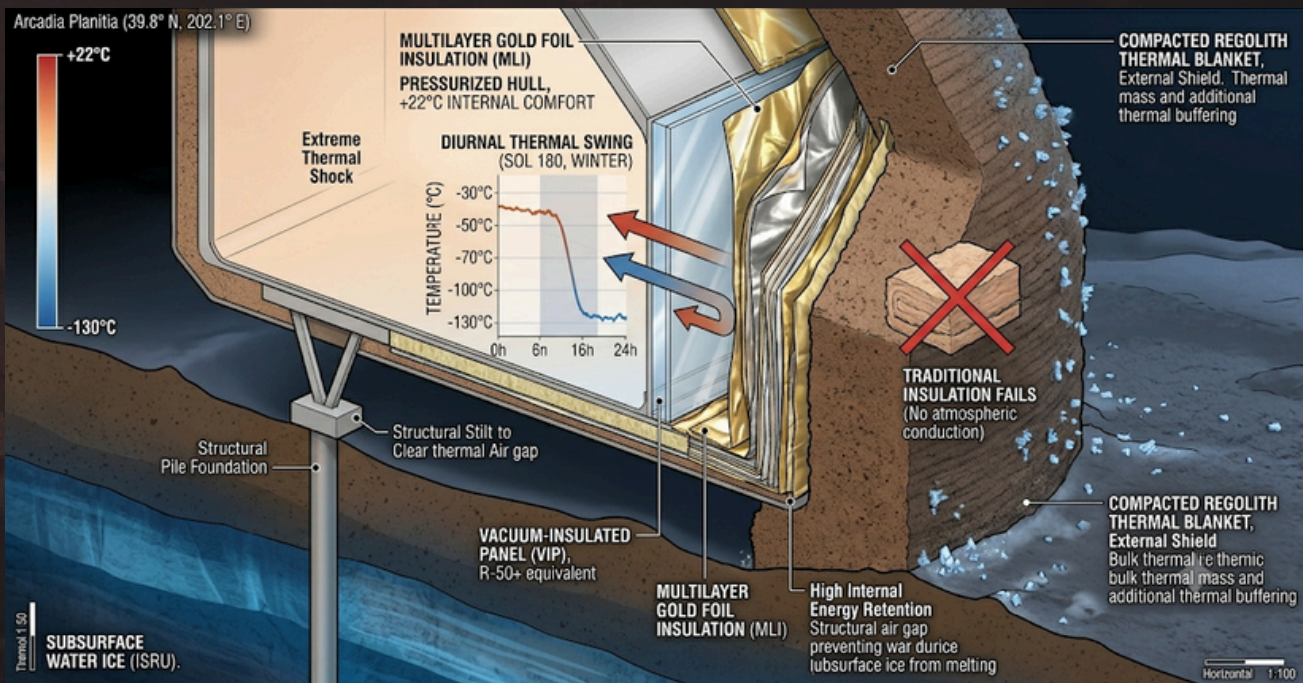


GREEN is the farming zone where are RED acts as the interactive zone

ENTRY



INSULATION



MIND

RAIN

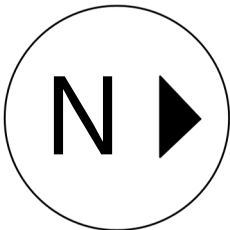
COMPITION
THE UNREAL
HOUSE

DESIGN MARTIAN HOUSE

SCALE 1:500

SHEETNO. 3

SHEET TITLE - SITE PLAN



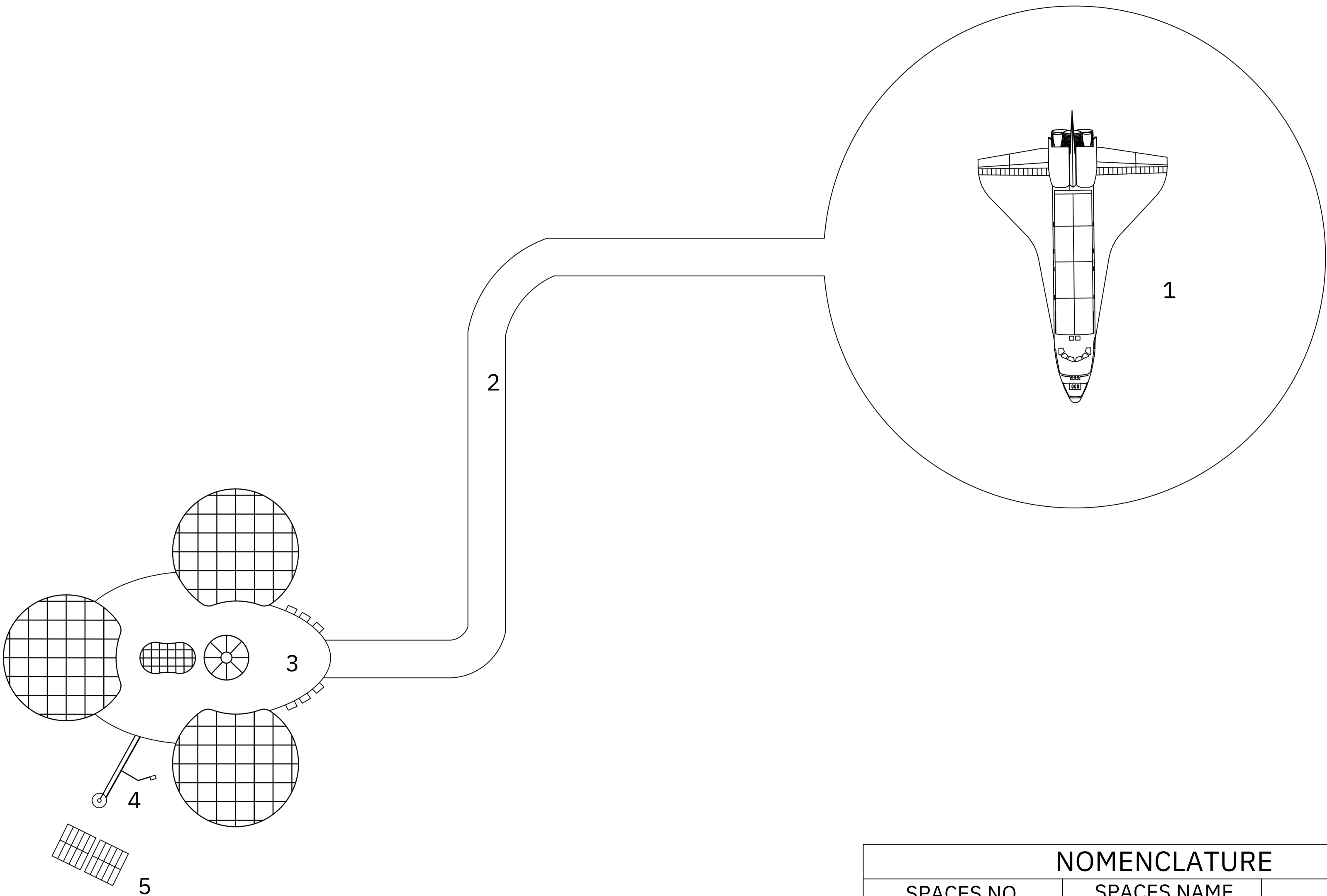
PARTICIPANTS -

GAURI VERMA

ANSHIKA

AYUSH KUMAR

COLLEGE:-
GATEWAY COLLEGE OF
ARCHITECTURE AND DESIGN,
SONIPAT , HARYANA



NOMENCLATURE

SPACES NO.	SPACES NAME	LEVEL
1	LANDING PAD	+150
2	ACCESS LANE	+150
3	RESIDENCE	+5800
4	WATER EXTRACTION WELL	+800
5	SOLAR PANEL	+1750 (at an angle of 39° towards south side)

MIND RAIN COMPITION

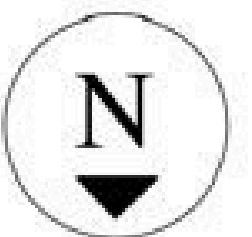
THE UNREAL HOUSE

DESIGN MARTIAN HOUSE

SCALE 1:150

SHEET NO.

SHEET TITLE PLAN



PARTICIPANTS -

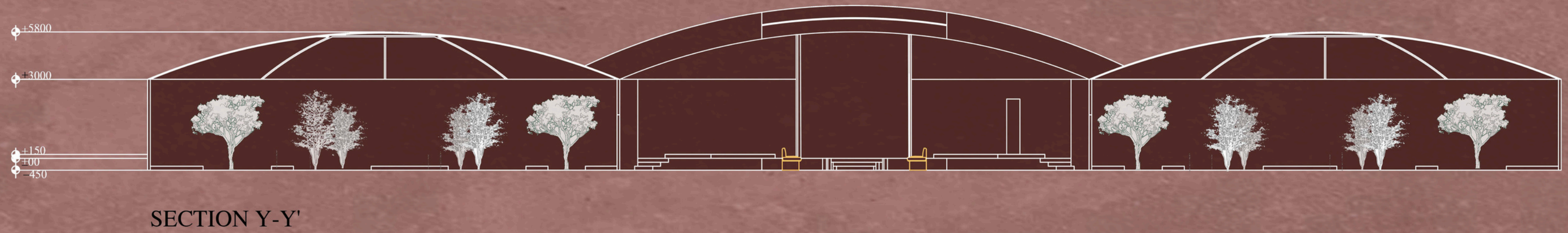
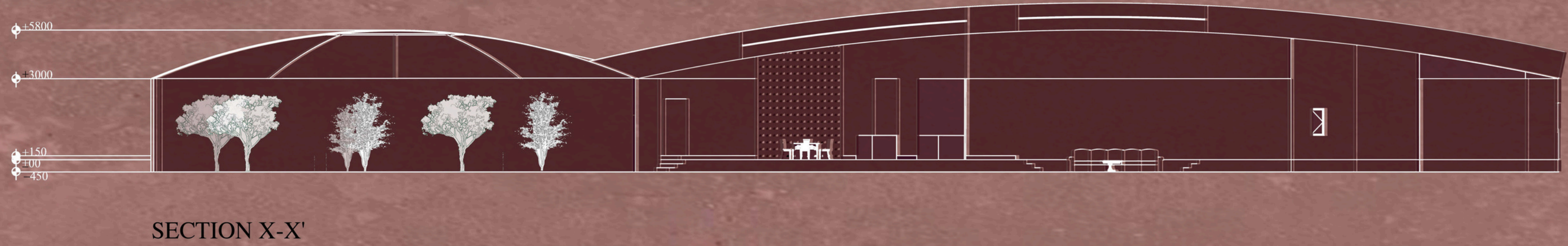
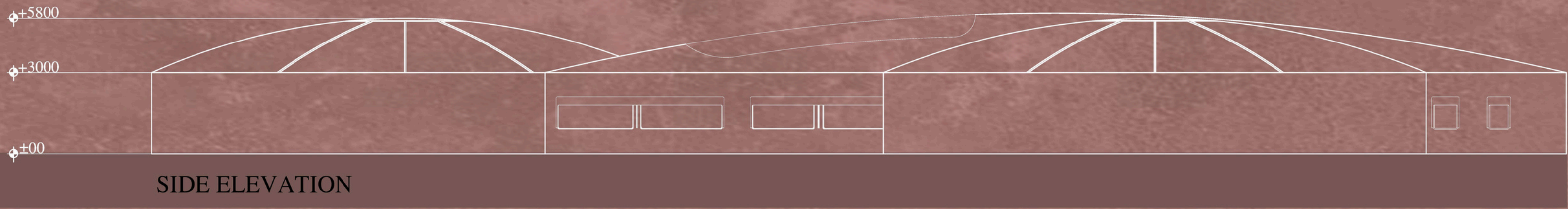
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GATEWAY COLLEGE OF
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MIND RAIN COMPITION

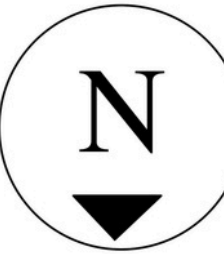
THE UNREAL HOUSE

DESIGN MARTIAN HOUSE

SCALE 1:150

SHEET NO. 5

SHEET TITLE :- ELEVATIONS
& SECTIONS



PARTICIPANTS -

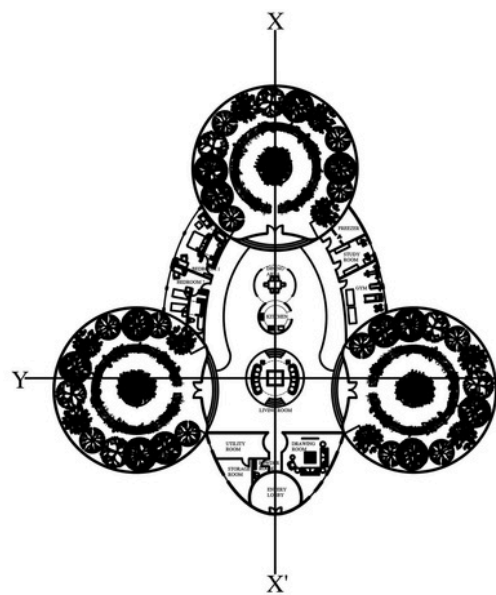
GAURI VERMA

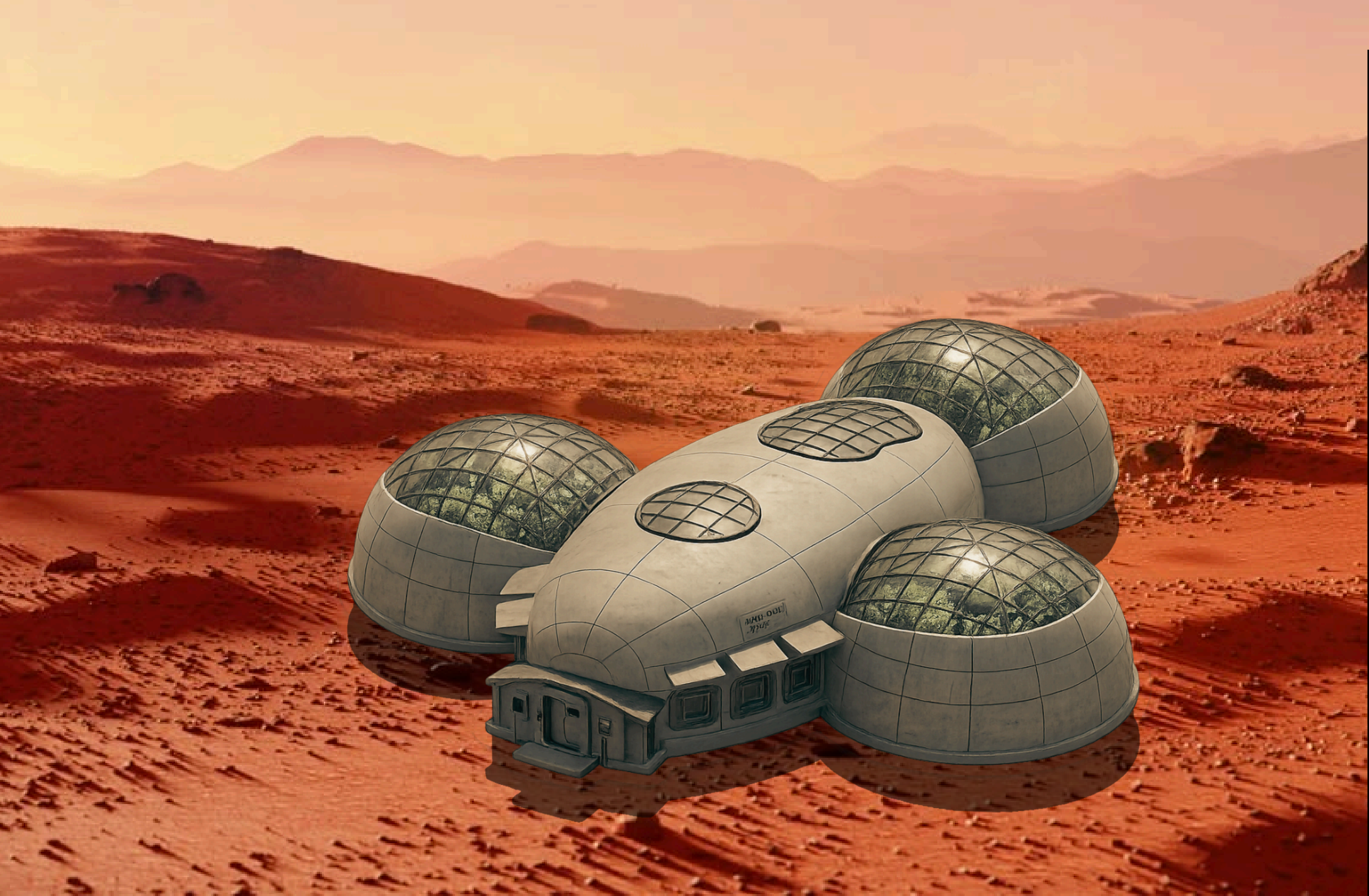
ANSHIKA

AYUSH KUMAR

COLLEGE:-
GATEWAY COLLEGE OF
ARCHITECTURE AND DESIGN,
SONIPAT , HARYANA

KEY MAP:-





MIND RAIN

COMPITION

THE UNREAL HOUSE

DESIGN MARTIAN HOUSE

SCALE

SHEET NO. 6

SHEET TITLE VIEWS

PARTICIPANTS-

GAURI VERMA

ANSHIKA KESHRI

AYUSH KUMAR

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